opposite charge. Depending on where the particles move towards, each pixel can have two states – one where the particles are towards the front of the display, (in which case the dense titanium oxide particles reflect the light back to the user and the pixel appears white), and one in which particles are towards the back of the display, (in which case the coloured dye comes in the way and absorbs the incident light and the pixel appears dark).

One more point of acceleration for this technology which brought it to its current state happened in the labs of MIT in the '90s with a team of undergraduates, who went on to start the E-Ink Corporation in 1997 to commercialize the technology. Some of the major issues with standard electrophoretic displays were their short lifetimes and the difficulty to manufacture the same. The approach that solved this problem was micro-encapsulation, a process which involves encapsulating a solid or a liquid in capsules of microscopic proportions to increase its durability as well as control the rate at which it tends to be released from the capsule over time. It also allows a process similar to printing to be used for manufacturing the displays, making it much more practical as an electronic analogue to paper.

An Alternative method – Electrowetting

The proper scientific definition of electrowetting process states that it is the modification of the wetting properties of a surface by applying electric voltage. For electronic displays, this technology is effectively put to use by controlling the shape of an encapsulated water-oil interface by applying voltage across electrodes.

In the absence of any external voltage, the oil water interface behaves normally with the oil forming a flat layer over the water and comes between the hydrophobic electrode surface and the water. Applying a certain charge to the encapsulation causes the stack to become unstable and changes the surface tension between the hydrophobic coating on the electrode and the water itself, causing the water to move towards the electrode and as a result the coloured oil is moved aside.

In the first state, any incoming light is faced by the coloured oil and as a result the absorbed light makes the pixel appear coloured. In the second state, the water makes the pixel partly transparent and possibly white, if there is an underlying reflective white surface.

Due to the small size of the pixel this type of a display offers a high-brightness and high-contrast switchable element, with switching speed between white and coloured reflection fast enough to display video content.